

Week 3 Lab - Control Flow & Functions

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Course: Introduction to Big Data Analytics

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```
import os
try:
    from google.colab import files
    uploaded = files.upload()
except ImportError:
    pass
```

[Choose Files](#) week3_students.csv

week3_students.csv(text/csv) - 2538 bytes, last modified: 9/21/2026 - 100% done
Saving week3_students.csv to week3_students (1).csv

Grading ladder

```
test_scores = [91, 76, 64, 52, 43]

for score in test_scores:
    if score >= 85:
        grade = "A"
    elif score >= 70:
        grade = "B"
    elif score >= 60:
        grade = "C"
    elif score >= 50:
        grade = "D"
    else:
        grade = "F"
    print(score, "->", grade)
```

```
91 -> A
76 -> B
64 -> C
52 -> D
43 -> F
```

Course validation

```
test_cases = [(76, 92), (76, 68), (43, 95)]

for score, attendance in test_cases:
    if score >= 50 and attendance >= 75:
        status = "Validated"
    elif score >= 50 and attendance < 75:
        status = "Blocked: low attendance"
    else:
        status = "Failed"
    print(f"score={score}, att={attendance}% => {status}")
```

```
score=76, att=92% => Validated
score=76, att=68% => Blocked: low attendance
score=43, att=95% => Failed
```

Loops

```
scores = [72, 85, 91, 64, 78, 47, 88, 55, 93, 61]

for score in scores:
    if score >= 80:
        print("Top score:", score)
```

```
Top score: 85
Top score: 91
Top score: 88
Top score: 93
```

```

total = 0
count = 0
passed = 0
failed = 0

for s in scores:
    total = total + s
    count = count + 1
    if s >= 50:
        passed = passed + 1
    else:
        failed = failed + 1

average = total / count
print("total:", total)
print("count:", count)
print("average:", round(average, 1))
print("passed:", passed)
print("failed:", failed)

```

```

total: 734
count: 10
average: 73.4
passed: 9
failed: 1

```

```

balance = 0
months = 0

while balance < 250000:
    balance = balance + 15000
    months = months + 1

print("months needed:", months)
print("balance:", balance)

```

```

months needed: 17
balance: 255000

```

▼ Functions

```

def get_grade(score):
    """returns letter grade for a score"""
    if score >= 85:
        return "A"
    elif score >= 70:
        return "B"
    elif score >= 60:
        return "C"
    elif score >= 50:
        return "D"
    return "F"

print(get_grade(91))
print(get_grade(76))
print(get_grade(43))

```

```

A
B
F

```

```

def pass_rate(score_list):
    """returns percentage of scores that are 50 or above"""
    passed = 0
    total = 0
    for s in score_list:
        total = total + 1
        if s >= 50:
            passed = passed + 1
    return (passed / total) * 100

print(pass_rate(scores))

```

```

90.0

```

▼ Reading the file

```
with open("week3_students.csv") as f:
    lines = f.readlines()

print("lines:", len(lines))
print("header:", lines[0].strip())
print("first student:", lines[1].strip())
```

```
lines: 41
header: student_id,name,age,gender,program,district,attendance_pct,score
first student: AUCA001,Aline Uwase,19,F,Information Systems,Gasabo,92,85
```

```
names = []
for line in lines[1:]:
    parts = line.strip().split(",")
    names.append(parts[1])
```

```
print(len(names))
print(names[:5])
```

```
40
['Aline Uwase', 'Eric Niyonzima', 'Diane Mukamana', 'Jean Bosco Habimana', 'Sandrine Ingabire']
```

✓ Pipeline

```
total = 0
valid = 0
bad = 0
bad_ids = []

for line in lines[1:]:
    parts = line.strip().split(",")
    try:
        score = int(parts[7])
        total = total + score
        valid = valid + 1
    except ValueError:
        bad = bad + 1
        bad_ids.append(parts[0])

average_valid = total / valid

print("valid:", valid)
print("bad:", bad)
print("bad ids:", bad_ids)
print("average:", round(average_valid, 1))
```

```
valid: 37
bad: 3
bad ids: ['AUCA008', 'AUCA020', 'AUCA032']
average: 75.3
```

```
passed = 0
failed = 0
best_score = -1
best_name = ""

for line in lines[1:]:
    parts = line.strip().split(",")
    try:
        score = int(parts[7])
        name = parts[1]
        if score >= 50:
            passed = passed + 1
        else:
            failed = failed + 1
        if score > best_score:
            best_score = score
            best_name = name
    except ValueError:
        pass

print("passed:", passed)
print("failed:", failed)
print("top student:", best_name)
print("top score:", best_score)
print("top grade:", get_grade(best_score))
```

```
passed: 35
failed: 2
top student: Bonaventure Nkurunziza
```

```
top score: 95
top grade: A
```

```
sums = {}
counts = {}

for line in lines[1:]:
    parts = line.strip().split(",")
    d = parts[5]
    try:
        s = int(parts[7])
        sums[d] = sums.get(d, 0) + s
        counts[d] = counts.get(d, 0) + 1
    except ValueError:
        pass

best_district = ""
best_avg = -1

for d in sorted(sums):
    avg = sums[d] / counts[d]
    print(d, "->", round(avg, 2))
    if avg > best_avg:
        best_avg = avg
        best_district = d

print("best district:", best_district, round(best_avg, 2))
```

```
Gasabo -> 80.4
Huye -> 68.4
Kicukiro -> 64.0
Muhanga -> 87.0
Musanze -> 65.2
Nyagatare -> 82.8
Nyarugenge -> 74.8
Rubavu -> 78.67
best district: Muhanga 87.0
```

Output file

```
with open("week3_report.txt", "w") as f:
    f.write("Week 3 report - Manzifred 26634\n")
    f.write("valid: " + str(valid) + "\n")
    f.write("bad: " + str(bad) + "\n")
    f.write("broken ids: " + str(bad_ids) + "\n")
    f.write("average: " + str(round(average_valid, 1)) + "\n")
    f.write("passed: " + str(passed) + "\n")
    f.write("failed: " + str(failed) + "\n")
    f.write("top student: " + best_name + " score=" + str(best_score) + " grade=" + get_grade(best_score) + "\n")
    f.write("best district: " + best_district + " avg=" + str(round(best_avg, 2)) + "\n")

with open("week3_report.txt") as f:
    print(f.read())
```

```
Week 3 report - Manzifred 26634
valid: 37
bad: 3
broken ids: ['AUCA008', 'AUCA020', 'AUCA032']
average: 75.3
passed: 35
failed: 2
top student: Bonaventure Nkurunziza score=95 grade=A
best district: Muhanga avg=87.0
```

```
try:
    from google.colab import files
    files.download("week3_report.txt")
except ImportError:
    pass
```

Reflection

At 7.5% skipping is fine, 37 records is still enough. But if it went up to 15 or 20 percent I would be more careful because the missing ones might all be from the same district and that would mess up the results. I would try to get the real scores first, if not possible I would fill them in with the district average and mark those rows as estimated.

Variable scope confused me. I created a variable inside a function and tried to use it outside and got a NameError. I didnt know variables inside a function disappear once it finishes. You have to use return to get the value out, once I got that it made sense.